# Probability basics: random variables, expectation, variance and covariance

*A chat model gives every possible next word a probability. What does that number mean, and why do they all add up to exactly one?*

Mathematics for AI and Machine Learning | Session 13 of 18 | Probability and statistics

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# Today in one view: what you will leave with

## By the end of today you can



Describe a sample space, an event and a random variable, and use the rules of probability on dice

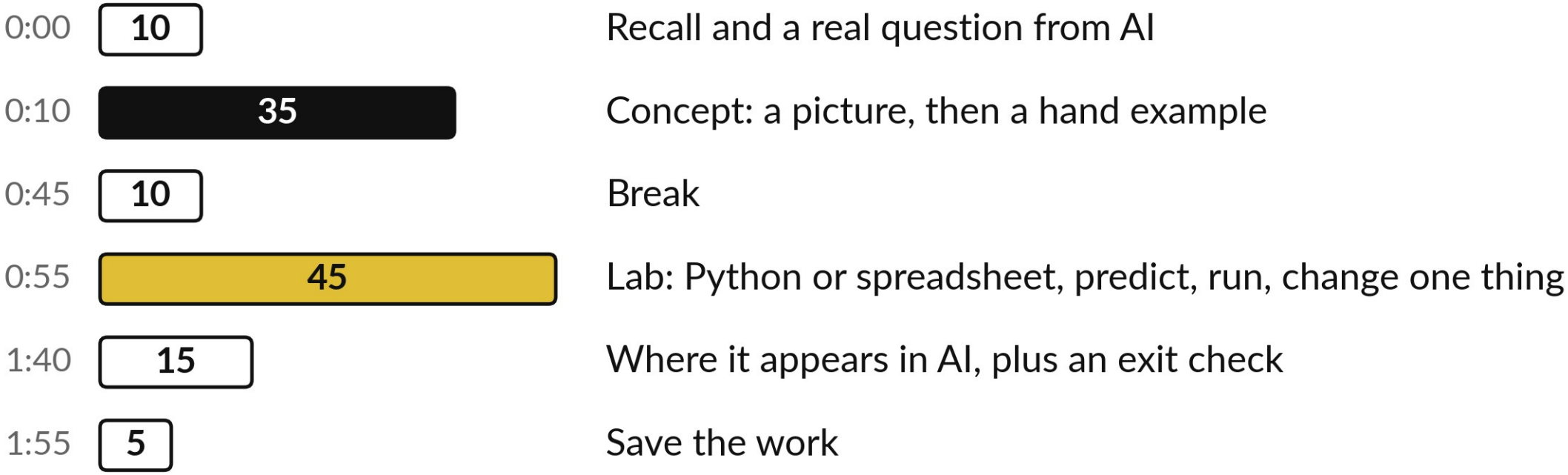


Calculate an expectation, a variance, a standard deviation, a covariance and a correlation by hand and by computer



Read a next-word probability list from a chat model as a probability mass function

# Your two hours: dice and data first, then the lab, then AI




Pictures and hand calculations first. Then the lab on dice, next-word probabilities and a small dataset.

# Three questions from last time

1. Why can computing e to the power of a large score break a softmax, and what does the stable softmax do about it?

2. What does regularisation (an L2 penalty, also called weight decay) add to the loss, and why?

3. Many models use 16-bit numbers. What is the saving and what is the risk?




Answer aloud first. Then we check together.

# Answers to the three questions from Session 12

| Question | Short answer |
|---|---|
| **Why can computing e to the power of a large score break a softmax, and what does the stable softmax do about it?** | The number overflows (too large for the computer). The stable softmax subtracts the largest score first. The probabilities stay the same. |
| **What does regularisation (an L2 penalty, also called weight decay) add to the loss, and why?** | A penalty that grows with the size of the weights. It keeps weights small and reduces overfitting. |
| **Many models use 16-bit numbers. What is the saving and what is the risk?** | Less memory and less time. The largest number in the 16-bit floating-point format fp16 is only 65,504, so overflow comes sooner (bfloat16 keeps the range but loses precision). |

If one answer surprised you, look back at Session 12 before the lab.

# Words we use today

| | Word | What it means |
|---|---|---|
|  | **Random variable** | A rule that turns the outcome of a chance event into a number. We write it X. |
|  | **Probability mass function (pmf) and density function (pdf)** | The pmf gives P(X = x) for discrete values (values you can list, such as the faces 1 to 6). The pdf is a curve, and probability is the area under it. |
|  | **Expectation** | The probability-weighted average of X, written E(X) or mu. |
|  | **Variance and standard deviation** | The average squared distance from mu, and its square root sigma. |
|  | **Covariance and correlation** | How two columns move together, and the same thing scaled to lie between -1 and +1. |

# A sample space lists every outcome, and an event is a group of outcomes

| Die 2 (rows) \ Die 1 (columns) | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **1** | 2 | 3 | 4 | 5 | 6 | **7** |
| **2** | 3 | 4 | 5 | 6 | **7** | 8 |
| **3** | 4 | 5 | 6 | **7** | 8 | 9 |
| **4** | 5 | 6 | **7** | 8 | 9 | 10 |
| **5** | 6 | **7** | 8 | 9 | 10 | 11 |
| **6** | **7** | 8 | 9 | 10 | 11 | 12 |

**Two fair dice give 36 equally likely outcomes. Each cell shows the sum.**

Sample space: the set of all possible outcomes.

Event: a group of outcomes we care about. Gold = the event "sum is 7".

P(sum is 7) = 6 / 36 = 1/6, about 0.167.

# Probability is a number from 0 to 1, and all the outcomes together add up to 1




**Probability P(A) measures how likely an event A is.**

0 means impossible. 1 means certain.

For equally likely outcomes: P(A) = outcomes in A divided by all outcomes.

The probabilities of all the outcomes add up to 1.

# The rules of probability, checked on dice

Range and total $$0 \le P(A) \le 1, \qquad P(\text{all outcomes}) = 1$$

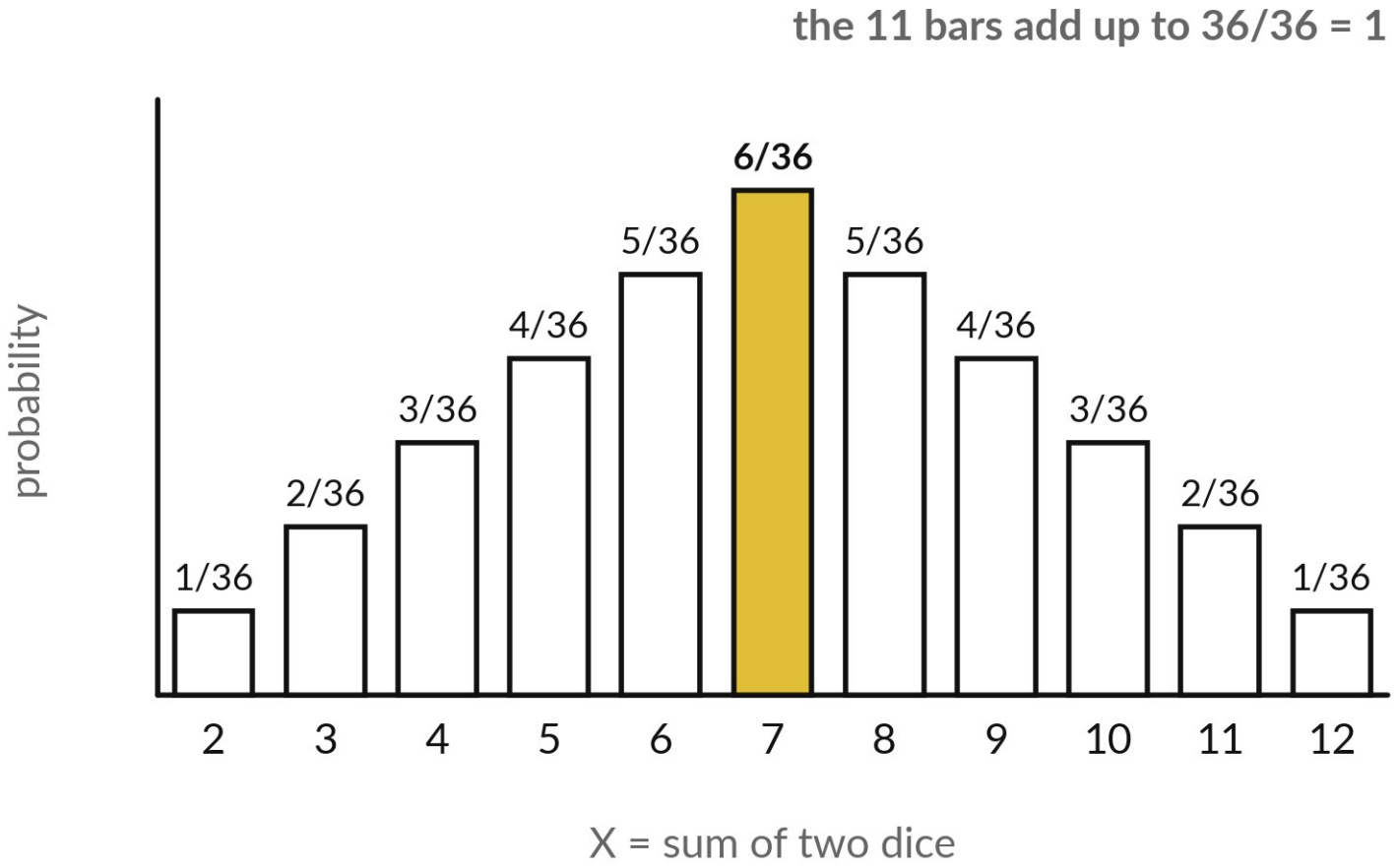
Complement $$P(\text{not } A) = 1 - P(A)$$

Either one $$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

| Symbol | What it means | In our data |
|---|---|---|
| **A, B** | Two events. A = the die is even {2, 4, 6}. B = the die shows more than 4 {5, 6}. | one fair die |
| **P(A), P(B)** | Their probabilities: 3/6 and 2/6. | P(A) = 0.5, P(B) = 0.333 |
| **A and B** | Both at once: only the face 6. | P(A and B) = 1/6 |
| **A or B** | At least one: 2, 4, 5, 6. The 6 is counted once, so we subtract it once. | 3/6 + 2/6 - 1/6 = 4/6 = 0.667 |

Complement example: P(not even) = 1 - 3/6 = 3/6.

# A random variable turns each outcome into a number, and its pmf says how likely each number is



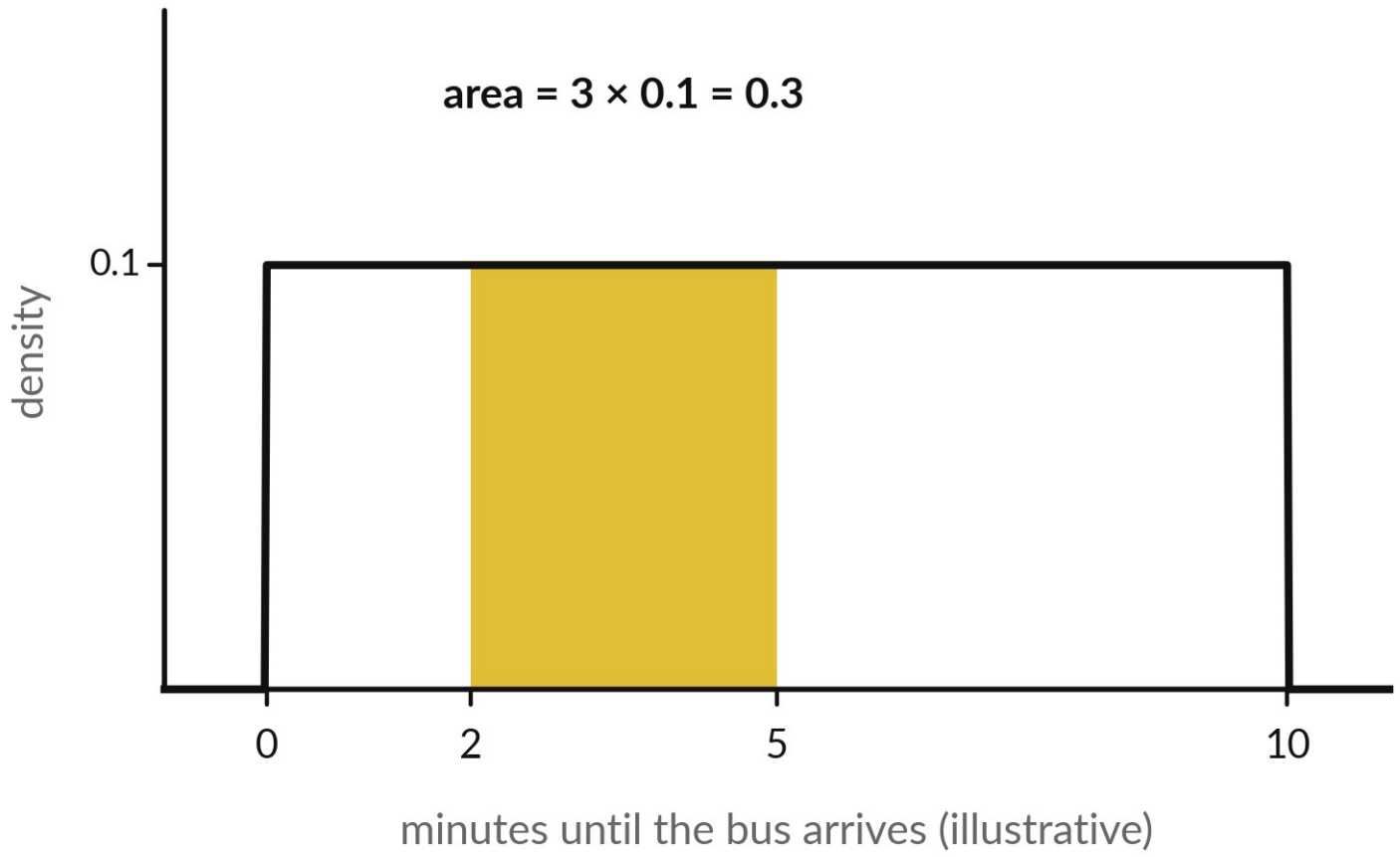

**Let X be the sum of two dice. The bar heights are the probability mass function (pmf): p(x) = P(X = x).**

A random variable X gives a number to each outcome.

P(X = 7) = 6/36 is the tallest bar (gold).

All the bars add up to 1.

# A continuous random variable has a density curve, and probability is the area under it




**A bus arrives at a random moment in a 10 minute window. This is a uniform distribution: every moment is equally likely.**

Density f(x) (pdf): the curve, flat at 1/10 = 0.1, so total area = 10 × 0.1 = 1.

P(2 to 5 minutes) = width 3 × height 0.1 = 0.3.

P(exactly 3.000...) = 0. Only ranges have probability.

# Probability mass function and density function, side by side

pmf (discrete X)

$$p(x) = P(X = x), \qquad \sum_x p(x) = 1$$

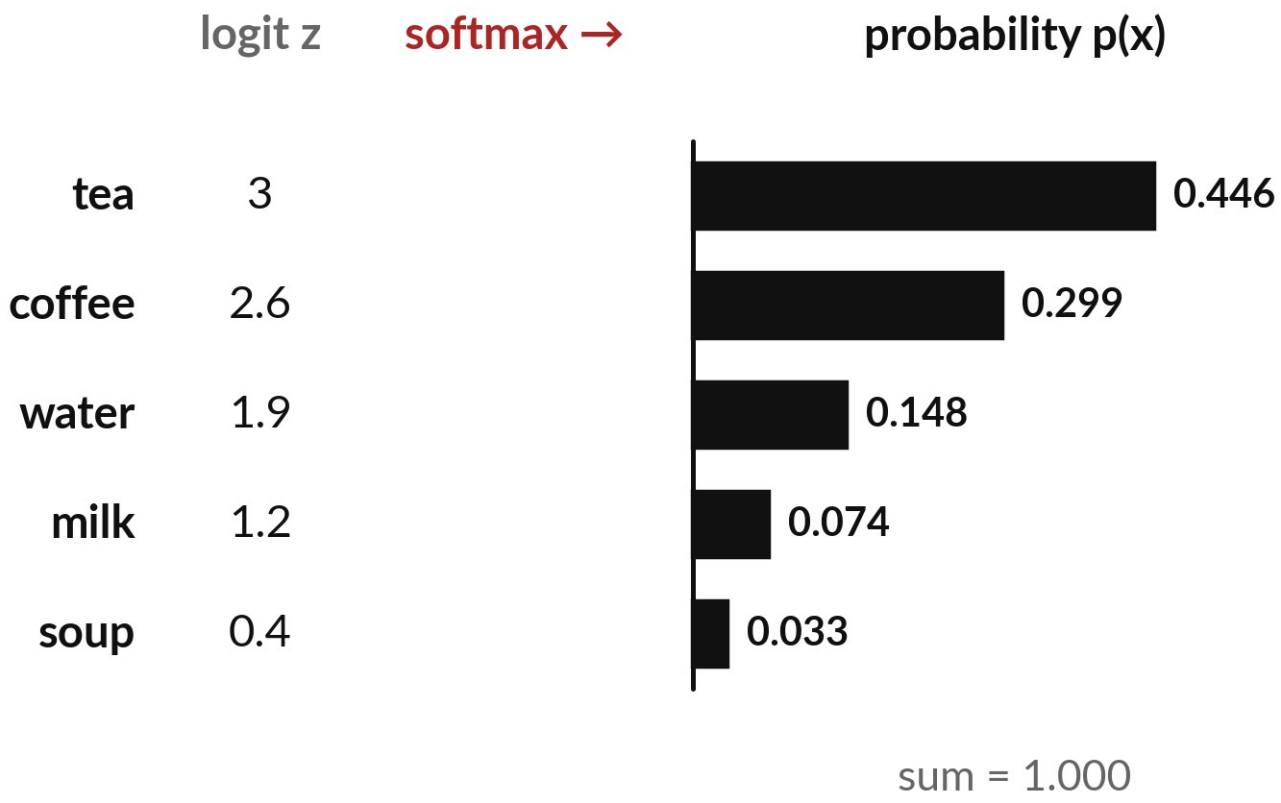
pdf (continuous X)

$$P(a \le X \le b) = \int_a^b f(x)\,dx, \qquad \int f(x)\,dx = 1$$

| Symbol | What it means | In our data |
|---|---|---|
| **p(x)** | The probability that X takes exactly the value x. Never negative. | die: p(x) = 1/6 |
| **f(x)** | The density: the height of the curve at x. It can be above 1. | bus: f(x) = 0.1 |
| **sum** | Add the probabilities of every possible value x. | 6 faces, total 1 |
| **integral** | The area under the curve from a to b. Here we read it as width times height. | 3 × 0.1 = 0.3 |

Both say the same thing: the total probability is 1. Discrete values use a sum, a continuous curve uses an area.

# A chat model gives every possible next word a probability, and the probabilities add up to 1



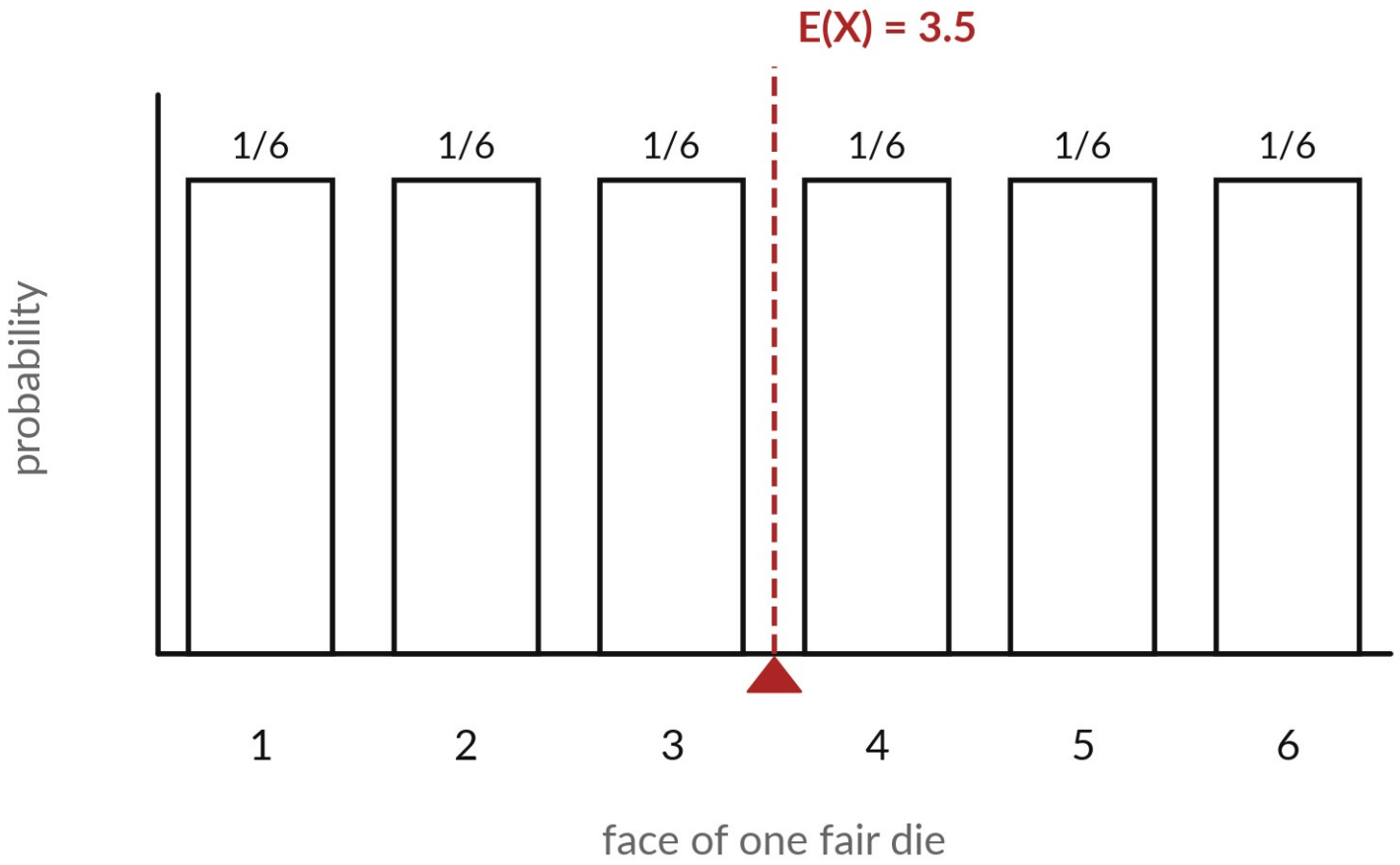

**Illustrative (made-up numbers): next word after "For breakfast I drink hot ...".**

Logits z: one raw score per word.

Softmax turns the scores into probabilities.

0.446 + 0.299 + 0.148 + 0.074 + 0.033 = 1.

# Expectation is the probability-weighted average, the balance point of the bars




**Expectation E(X), also called the mean and written mu, is the value the average settles on over many repeats.**

Picture the bars as weights on a ruler. E(X) is where it balances.

It need not be a possible value: no die shows 3.5.

# Expectation: add up each value times its probability

Expectation

$$E(X) = \mu = \sum_x x\, p(x)$$

Fair die

$$E(X) = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + \cdots + 6 \cdot \tfrac{1}{6} = \tfrac{21}{6} = 3.5$$

| Symbol | What it means | In our data |
|---|---|---|
| **E(X), mu** | The expectation (mean) of the random variable X. | 3.5 |
| **x and p(x)** | One possible value of X, and its probability. | face 1 to 6, 1/6 each |
| **sum** | Add the products x times p(x) over all values. | six terms |

Next-word example (illustrative): lengths 3, 6, 5, 4, 4 letters give E(length) = 4.30 letters.

# Variance and standard deviation measure how far values lie from the mean

Variance

$$\mathrm{Var}(X) = \sigma^2 = \sum_x (x - \mu)^2 \, p(x)$$

Standard deviation

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

Shortcut

$$\mathrm{Var}(X) = E(X^2) - \mu^2$$

| Symbol | What it means | In our data |
|---|---|---|
| **Var(X)** | Variance: average squared distance from mu (units of X squared). | 2.9167 |
| **sigma** | Standard deviation: the square root, in the units of X. | 1.7078 |
| **x - mu** | How far one value lies from the expectation. | for 6: 6 - 3.5 = 2.5 |
| **E(X^2)** | The expectation of X squared: sum of x squared times p(x). | die: 15.1667 - 12.25 = 2.9167 |

We square the distances so that below-mean and above-mean values do not cancel.

# By hand: the variance of a fair die is 35/12, about 2.92

| | A | B | C |
|---|---|---|---|
| | **x** | **x - 3.5** | **(x - 3.5) squared** |
| 1 | 1 | -2.5 | 6.25 |
| 2 | 2 | -1.5 | 2.25 |
| 3 | 3 | -0.5 | 0.25 |
| 4 | 4 | +0.5 | 0.25 |
| 5 | 5 | +1.5 | 2.25 |
| 6 | 6 | +2.5 | 6.25 |

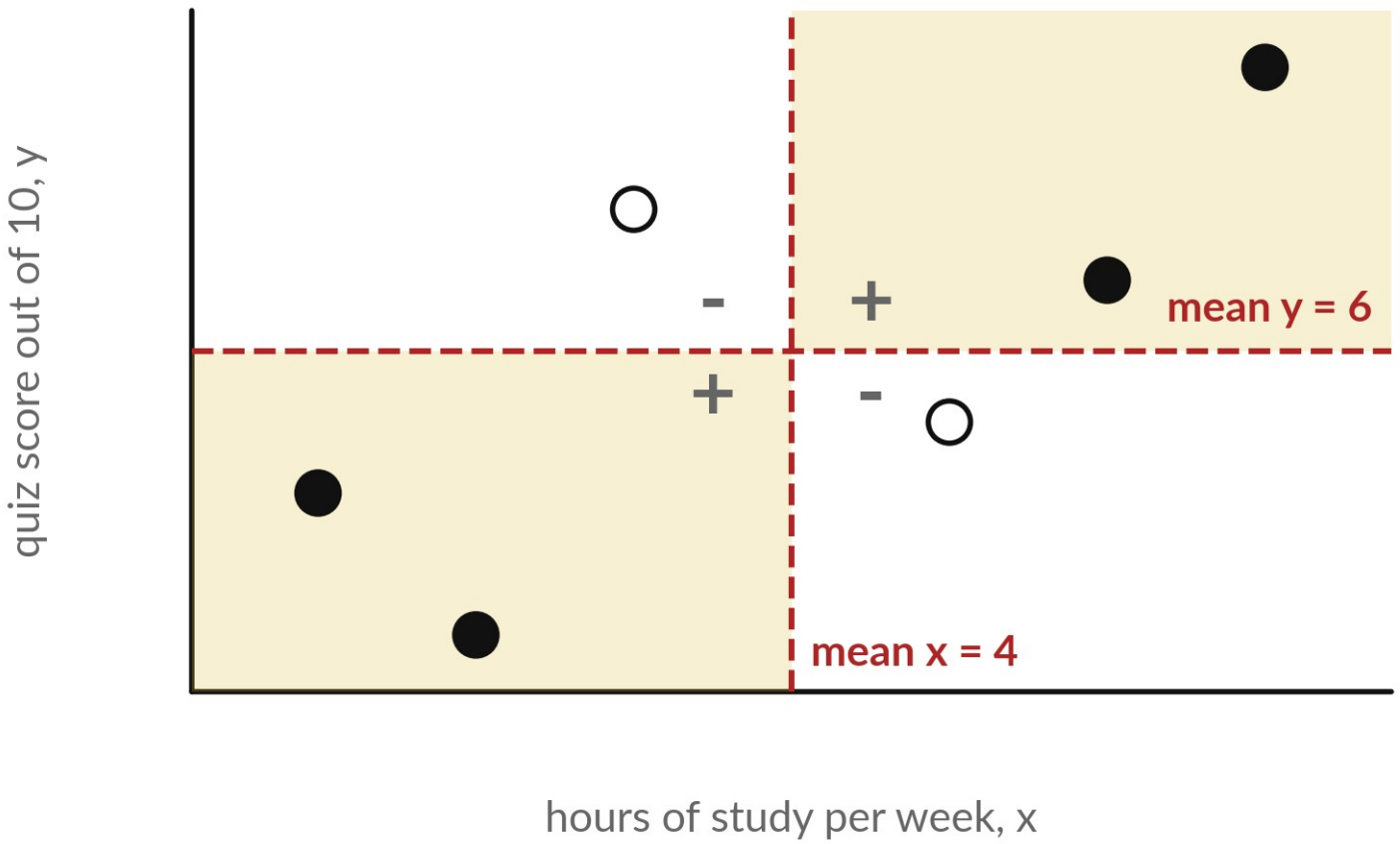
**Each face has p(x) = 1/6, so add the last column and divide by 6.**

Sum of the last column = 17.50.

Variance = 17.5 / 6 = 2.9167.

Standard deviation = square root = 1.7078.

# Covariance asks whether two columns rise and fall together




**Illustrative data: six learners. Dashed red lines mark the two means. Shaded corners are marked +.**

Dark dot: both above or both below the means. Product positive.

Open dot: product negative.

Per learner: (x - mean of x) times (y - mean of y). Covariance = mean of these products.

# By hand: the covariance of the six learners is 25/6, about 4.167

| | A | B | C | D | E |
|---|---|---|---|---|---|
| | x | y | x - 4 | y - 6 | product |
| 1 | 1 | 4 | -3 | -2 | +6 |
| 2 | 2 | 2 | -2 | -4 | +8 |
| 3 | 3 | 8 | -1 | +2 | -2 |
| 4 | 5 | 5 | +1 | -1 | -1 |
| 5 | 6 | 7 | +2 | +1 | +2 |
| 6 | 7 | 10 | +3 | +4 | +12 |
| 7 | | | | sum | 25 |

**The means are 4 hours and 6 points. Multiply the two distances for every learner.**

Covariance = 25 / 6 = 4.1667.

Variance of x = 4.6667.
Variance of y = 7.

Correlation = 4.1667 / (2.160 × 2.646) = 0.729.

# Covariance and correlation: how two columns move together

Covariance

$$\mathrm{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

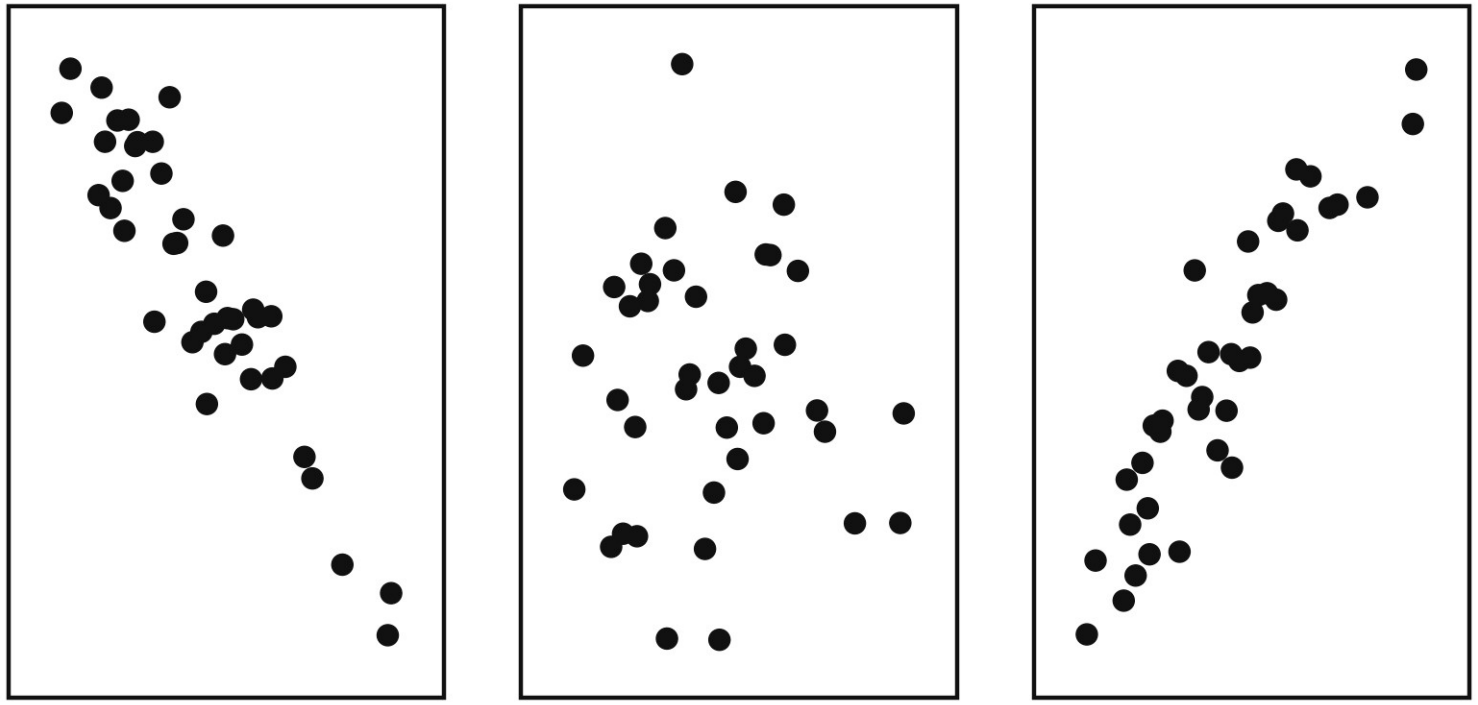
Correlation

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

| Symbol | What it means | In our data |
|---|---|---|
| **Cov(X, Y)** | The mean of the products of the two distances. | 4.1667 (hours × points) |
| **x-bar, y-bar** | The mean of each column. | 4 and 6 |
| **sigma X, Y** | The two standard deviations. | 2.160 and 2.646 |
| **r** | Covariance divided by both standard deviations. No units, between -1 and +1. (rho: same, for a population.) | 0.729 |

This course divides by n (population form). Cov of a column with itself is its variance.

# Correlation runs from -1 to +1, and it only sees straight-line patterns



r = -0.92

falls together

r = -0.02

no pattern

r = +0.92

rises together

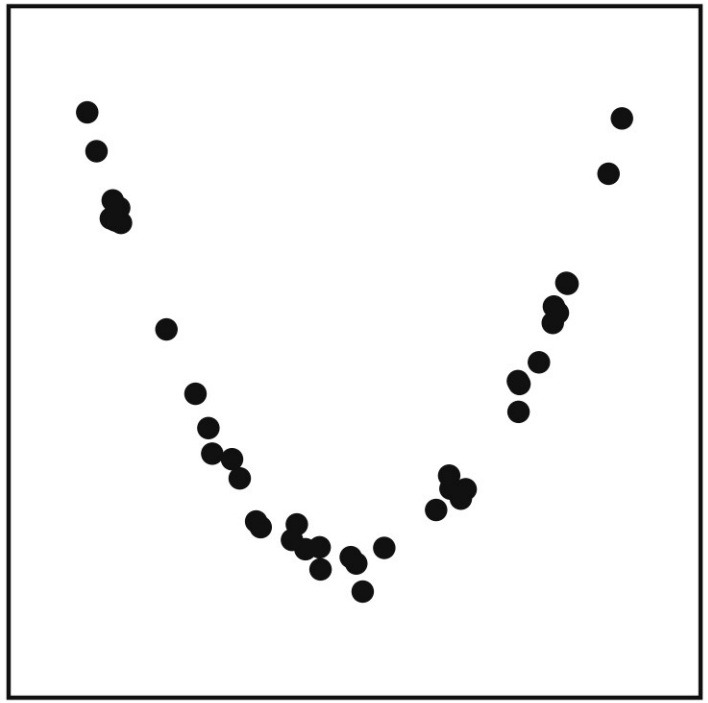
**Each panel shows 40 made-up points and their correlation r.**

r near +1: the columns rise together.

r near -1: one rises as the other falls.

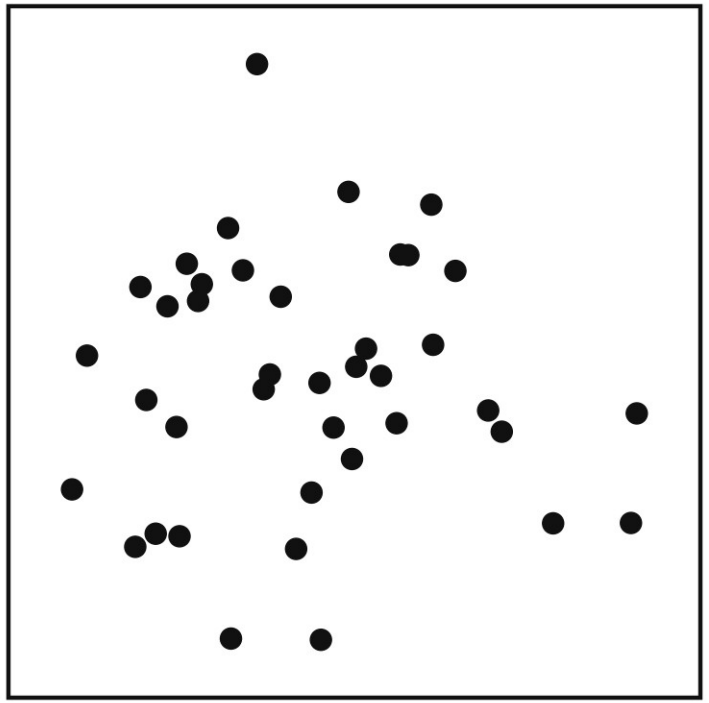
r near 0: no straight-line trend.

# Common mistake: a correlation near 0 does not mean the columns are unrelated



r = -0.10

clear U shape



r = -0.02

really no pattern

**Both panels have r close to 0. Only the right one has no relationship.**

Correlation measures straight-line trends only.

A strong curve can still give r near 0.

High correlation does not prove that one column causes the other.

# Two versions of variance: divide by n or divide by n minus 1

## Population: divide by n

VAR.P, COVARIANCE.P

np.var, np.cov(bias=True)

This course uses this version

## Sample: divide by n - 1

VAR.S, COVARIANCE.S

pandas .var(), .cov()

Used to estimate from a sample

**Divide by n: variance of y 7, covariance 4.167. By n - 1: 8.4 and 5.**

# Bridge: expectation, variance, covariance and correlation in NumPy

```
import numpy as np
x = np.array([1, 2, 3, 5, 6, 7])
y = np.array([4, 2, 8, 5, 7, 10])
np.mean(x), np.var(x), np.std(x)
np.cov(x, y, bias=True)[0, 1]
np.corrcoef(x, y)[0, 1]
```

## In plain words

Lines 2 and 3: the two columns as arrays.

Line 4: mean, variance and standard deviation. np.var divides by n.

Line 5: np.cov gives a 2 by 2 matrix. bias=True divides by n. The entry in row 0, column 1 is Cov(x, y).

Line 6: np.corrcoef gives the correlation matrix. Same entry is r.

Spreadsheet: AVERAGE, VAR.P, STDEV.P, COVARIANCE.P and CORREL do the same.

# Bridge: simulate dice with a fixed seed so everyone gets the same rolls

```
rng = np.random.default_rng(13)
rolls = rng.integers(1, 7, size=60)
rolls.mean(), rolls.var()
big = rng.integers(1, 7, size=10000)
big.mean(), big.var()
```

## In plain words

Line 1: a random number generator. Seed 13 fixes the sequence.

Line 2: 60 whole numbers from 1 to 6 (7 is excluded).

Line 3: 60 rolls give a noisy mean (here 3.9667) and variance (3.1322).

Lines 4 and 5: 10,000 rolls come much closer to 3.5 and 2.9167.

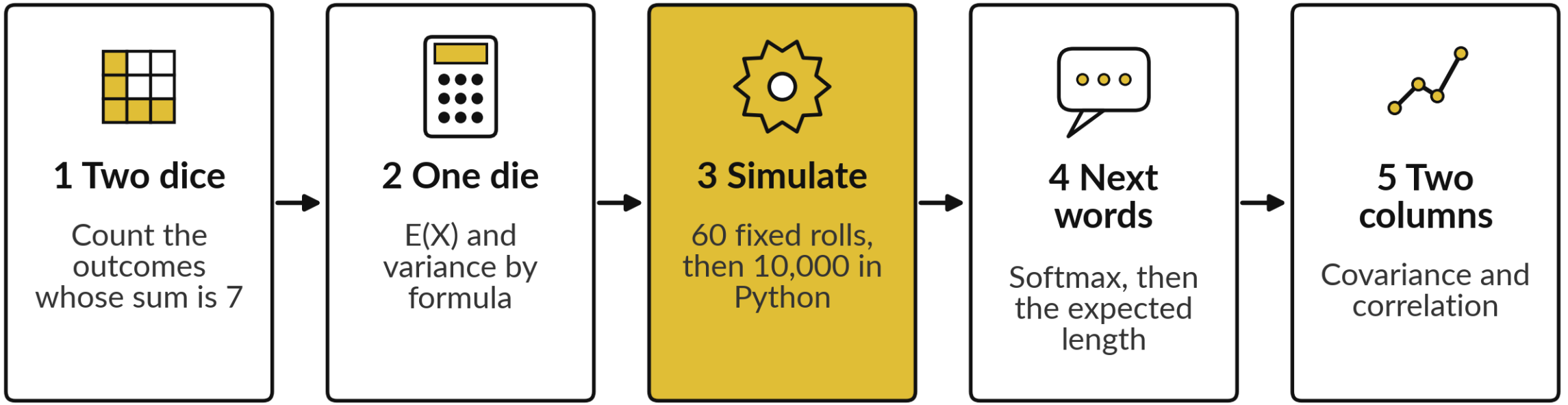
Spreadsheet: the 60 rolls are fixed values, so nothing changes.



# Part 2: the lab

Close the slides. Open the notebook or the workbook.

# The lab has five stages, in Python and in the spreadsheet




Python: notebook "Session 13: Probability basics". Spreadsheet: workbook with the sheets Dice, Die, Words and Covariance. Predict first, run, read, then change one thing.

# Check your numbers against these before you move on

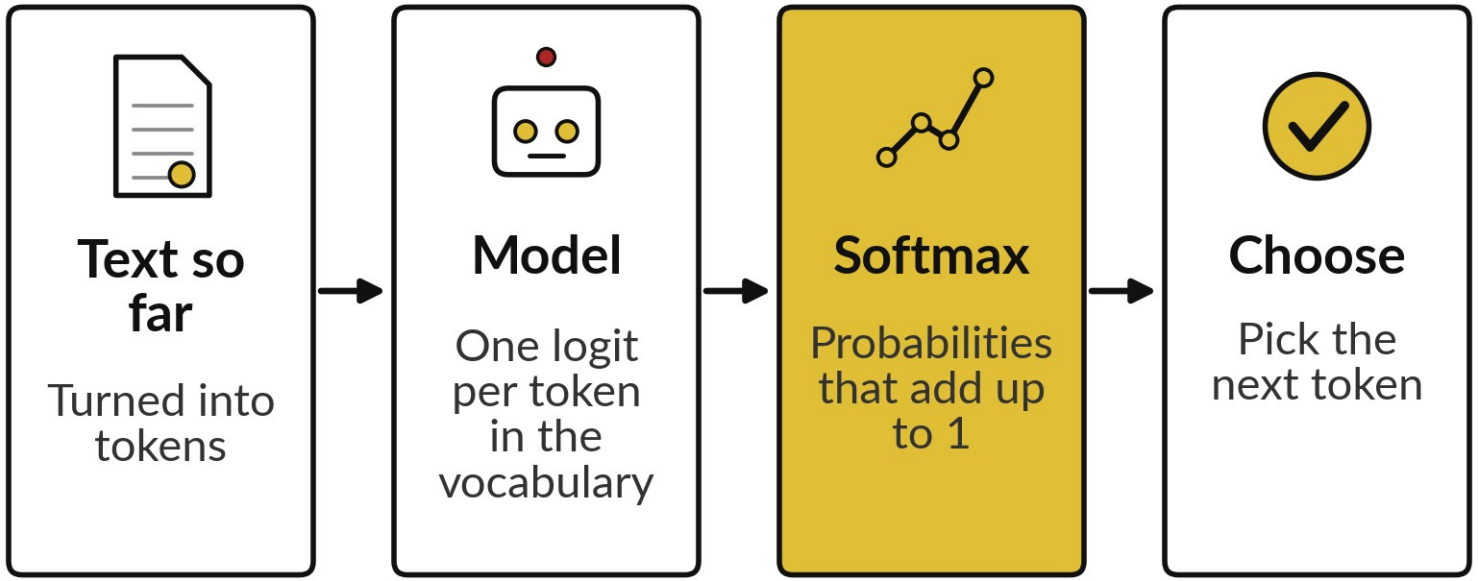
| What you compute | You should see |
|---|---|
| **Two dice: outcomes with sum 7, and P(sum is 7)** | 6 of 36, P = 0.1667 |
| **One fair die: E(X), variance, standard deviation** | 3.5, 2.9167, 1.7078 |
| **The 60 fixed rolls: mean and variance** | 3.9667 and 3.1322 |
| **Next words: P(tea), and the expected length in letters** | 0.4459 and 4.3003 |
| **x and y: covariance, variance of x, variance of y** | 4.1667, 4.6667, 7 |
| **x and y: correlation** | 0.7290 |
| **CHANGE task: covariance dividing by n - 1** | 5 |

If your number differs, stop and find the step that went wrong. Do not change a number to match.

# Errors you will meet, what they mean and what to do

| | Where | What you see | What to do |
|---|---|---|---|
| ✗ | **Python** | NameError: name x is not defined | A cell above was not run. Use Run all from the top. |
| ✗ | **Python** | Covariance 5.0 instead of 4.1667 | You divided by n - 1. Add bias=True to np.cov. |
| ✗ | **Spreadsheet** | Result 5 instead of 4.1667 | You used COVARIANCE.S. Use COVARIANCE.P. |
| ✗ | **Spreadsheet** | #NAME? in a formula | A function name is misspelled, or the sheet needs the English function names. |
| ✓ | **Both** | The 60 rolls give 3.9667, not 3.5 | Correct. A small sample is noisy. Only many rolls approach 3.5. |

# Where this appears in AI: a language model gives every next token a probability



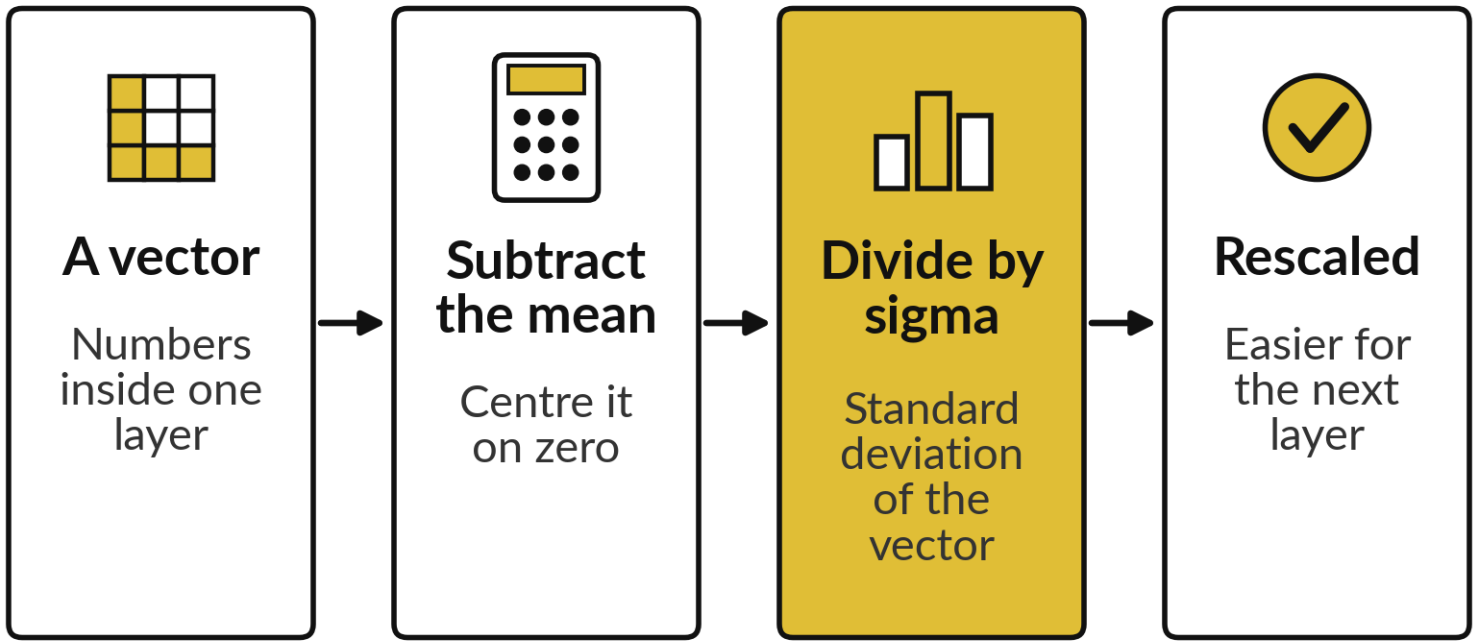

**Language models of this kind output a probability mass function over the whole vocabulary.**

Published transformers end in a softmax over the vocabulary.

Llama 2 (open): 32,000 tokens in the vocabulary.

Closed products do not publish details.

# Where this appears in AI: mean and standard deviation rescale numbers inside the model




**The original transformer uses layer normalisation: each vector is centred and scaled with its own mean and standard deviation.**

Layer normalisation: Ba, Kiros and Hinton, 2016.

Some newer open models use a variant (RMSNorm) that scales without centring.

# Where this appears in AI: check an assistant with your numbers

8 minutes

Check an AI assistant

## Ask an AI assistant for the covariance and correlation of your six pairs.

Type x = 1, 2, 3, 5, 6, 7 and y = 4, 2, 8, 5, 7, 10.

Ask for the covariance, the correlation and the divisor.

Compare with 4.167 and 0.729.

If it says 5, it divided by n - 1.

Never paste confidential data.

**You keep**

A note: its value, your value, match or not.

# Exit check: three questions

| Question | Short answer |
|---|---|
| **A model gives probabilities 0.5, 0.3 and 0.1 to three words and nothing else is possible. What is wrong?** | They add to 0.9. A pmf must add to exactly 1. |
| **What is the expectation of a fair die, and can you roll it?** | 3.5. No: it is an average, and no face shows 3.5. |
| **Two columns have correlation 0.02. Are they unrelated?** | Not necessarily. Correlation only sees straight-line trends. Draw the scatter plot. |

# Save the lab and keep your five numbers

## What you keep from today

Keep your saved notebook or workbook, and the five numbers from the check table: E(X) = 3.5, variance 2.9167, covariance 4.1667, correlation 0.729 and the next-word probabilities that add to 1.

### Our question today

*A chat model gives every possible next word a probability. What does that number mean, and why do they all add up to exactly one?*

### Next session

Session 14: distributions and sampling. We meet the Gaussian, the central limit theorem, an A/B test, and temperature and top-p sampling.

**Take away**

**Saved lab**
*Notebook or workbook*

**Five numbers**
*See the check table*

**Next session**
*Distributions and sampling*

**Bring**
*One question about chance*